UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING
04 OCT 2022
EXAMINATION & REGISTRATION

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UNIVERSITY OF COLOMBO, SRI LANKA

UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination – Semester I –2022

SCS-1201 – Data Structures and Algorithms I

TWO (2) HOURS

201

Answer ALL Questions

To be completed by the candidate

Examination Index No: .....

Important Instructions to candidates:

1. Students should answer in the medium of **English language only** using the space provided in this question paper.
2. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
3. Write your index number **CLEARLY** on each and every page of this Question paper.
4. This paper consists of **4** questions in **12** pages (including the Cover Page).
5. Programmable Calculators or any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are **not allowed**
6. **Non-Programmable** calculators are allowed.
7. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate
8. The paper will total 100 marks (***25 for questions 1 and 4, 30 marks for question 2 and 20 marks for question 3***).

For Examiner's use only	
Question No	**Marks**
1	
2	
3	
4	
Total	

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Question 1

(a) Consider the following infix expression.

3+4*(5*3-6)

(i) Write the resultant postfix expression of the above if one uses the usual Stacks algorithm to convert the infix to postfix.

(4 Marks)

(ii) When one evaluates the postfix expression obtained from a(i), what is the **maximum number** of tokens that will appear on the stack **at a time** during the evaluation of the expression?

(4 Marks)

(iii) What is/are the stack's remaining content(s) after evaluating the expression?

(4 Marks)

(iv) Consider that a multiplication operator (*) is added to the end of the original expression. When evaluating the expression after converting it into postfix, what would be the last stack content and the buffer input token value?

(5 Marks)

(b) (i) If one evaluates the following pseudocode algorithm segment using a dry run (hand execute), what would be the final content of the queue?

```
q = queue();
for(j=1; j<5;j++)
{
    q.enqueue (j);
}
for(i=1;i<5;i++)
{
    if(i%2 == 0)
    {
    q.dequeue()
    }
    else{
    q.enqueue(i*4)
    }
}
```

(4 Marks)

(ii) Hand execute (dry run) the following pseudocode segment and write the final content of the resulting stack.

(4 Marks)

```
stack=[]
for  i = 1 to 12
  if i%4 <> 0 then
   stack.append( i )
  else
    if i%2=0 then
      stack.pop()
    endif
  endif
next i
```

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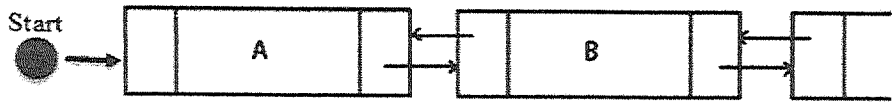
Question 2

(a) Consider the following straight doubly linked list with the initial pointer "***Start***".

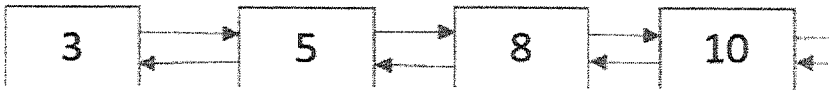


(i) Write a pseudocode or C code algorithm to remove Node B from the above List.

Note: (I) You may assume forward and backward reference as **next** and **prev** respectively.
(II) Do not use additional pointers other than the given pointes.

(2 Marks)

(ii) Write a pseudo code algorithm to insert a new node into a sorted linked list whilst maintaining the sorted order. An example test case is given below.
As a test case, when you test the algorithm, you may use the content of the new node as nine (09) and the initial list as given below.



After inserting the new node (09), the final sorted doubly linked list is as follows:

(4 Marks)

(iii) A polynomial p(x) is the expression in variable x which is in the form of $ax^n + bx^{n-1} + \ldots + jx + k$, where a, b, c ...., j, k fall in the category of real numbers and 'n' is a non-negative integer

Describe, how to represent the above polynomial using a singly linked list.

(2 Marks)

(iv) Consider the following two polynomials:

Polynormal 1 :$5x^2 + 4x^1 + 2x^0$
Polynormal 2 :$5x^1 - 5x^0$

Using the features of the linked lists, write a pseudocode algorithm to subtract polynomial 2 from polynomial 1.

(4 Marks)

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(b) The Fibonacci numbers are a sequence of integers in which the first two elements are 0 & 1, and each following element is the sum of the two preceding elements:

(i) Write a formula to generate the nth Fibonacci number.

(2 Marks)

(ii) Write a recursive pseudocode algorithm to find the nth Fibonacci number

(3 Marks)

(iii) Prove the time complexity of the recursive Fibonacci algorithm as O(2^n)).

(4 Marks)

(iv) Consider the following relationship related to the running time of the algorithm.

$T(n)=4n^3+n^2+2n+5$

Prove the complexity of the algorithm as $O(n^3)$ using Big-Oh Rules.

(3 Marks)

(v) Which parameters best describe the criteria for comparing the efficiency of algorithms? Justify the answer.

(3 Marks)

(vi) Consider following time complexity of algorithms.as any input of size n:

$T(n) = n^2+3n+5$

"When estimating running time of a relation for larger n values, there is not significant different when considering all the terms in the relation or considering the most dominant component of the relation"

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Prove the validity of the above expression by substituting different values for n in the above relationship.

(3 Marks)

Question 3

(a) One wants to create a max-heap using the following set of integers.

[6,2,14,1,7,21]

Nodes are inserted one by one and any heap property violation is fixed when detected.

Which of the following are valid/invalid statements?

(i) During the max heap creation, 14 is swapped with 6
(ii) The total number of swaps during the heap creation is 5.
(iii) When creating the max heap, 21 is swapped with 6, and again 21 is swapped with 14.
(iv) At the intermediate stage of heap construction, 14 is a root value.
(v) 2 never becomes a root value during the heap construction

(5 Marks)

(i).....
(ii).....
(iii).....
(iv).....
(v).....

(b) Consider the following two techniques when selecting a pivot value for partitioning a file in a quick sort algorithm.

- Pick the first element as the pivot.
- Pick median as the pivot.

Compare the suitability of the above techniques when one applies the partitioning algorithm for the following cases.

(i) All the elements in the file are already sorted in increasing order.
(ii) The elements in the file are arranged in random order.

(5 Marks)

(i)	(ii)

(c)Suppose one sorts an array of ten integers using quicksort, and if he/she has just finished the first partitioning as follows:

16,23,40,57,89,90,75,92,67,100

What are all the possible pivot values that can choose to obtain the above intermediate output?

(5 Marks)

(d) Consider the following straight merge sort algorithm

Step 1 : Divide the file into n sub files of size 1

Step 2: merge adjacent pairs of files according to their values in increasing order (Then we have approximately n/2 files of size 2)

Step 3: Repeat this process (step 1 and step 2) until there is only one file remaining of size n.

If one uses the above straight sort algorithm to sort the following data set, how many passes(rounds) are required to completely sort the data set

Data set is: 25,57,48,37,12,92,86,33

(5 Marks)

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Question 4

(a) (i) The output of the level order traversal is given below:

7, 4, 12, 3, 6, 8, 1, 5, 10

Consider the following algorithm

Step 1: First pick the first element of the array and make it root.
Step 2: Pick the second element, if its value is smaller than root node value makes it left child,
Step 3: Else make it right child

The recursive call Step 2 and Step 3 continued to make the tree until all the nodes are picked.

What would be the resultant tree if one applies the above data set to the algorithm?

(4 Marks)

(ii) What is the minimum and the maximum number of nodes in an AVL tree of height 3 and 4 respectively?

Hint: The height of a tree consisting of single node is 0, Give an exact number for the answer. Do not provide the formula.

(4 Marks)

	Height=3	Height=4
Max. number of Nodes		
Min. Number of Nodes		

(iii) What is the Maximum and the minimum height of an AVL tree when the total number of nodes is equal to 8, 10 and 13 respectively

(4 Marks)

No. of Nodes	Maximum Height	Minimum Height
8		
10		
13		

(b) (i) Draw the weighted directed graph represented by the adjacency matrix below. A nonzero value at [row, column] indicates that the vertex in the row is adjacent to the vertex in the column:

(3 Marks)

	P	Q	R	S	T
P	1	0	0	1	1
Q	1	0	1	0	0
R	1	0	0	0	1
S	1	1	0	0	0
T	1	0	0	1	0

(ii) Describe how one can find the path matrix using the given adjacency matrix.

(4 Marks)

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(iii) Write the path matrix according to the methodology described in part b(ii)) above

(3 Marks)

(iv) Describe the Breadth-First Traversal of a graph proposed by you in part b(i) above using a queue.

(3 Marks)

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